EXAMPLE: BIVARIATE GAUSSIAN

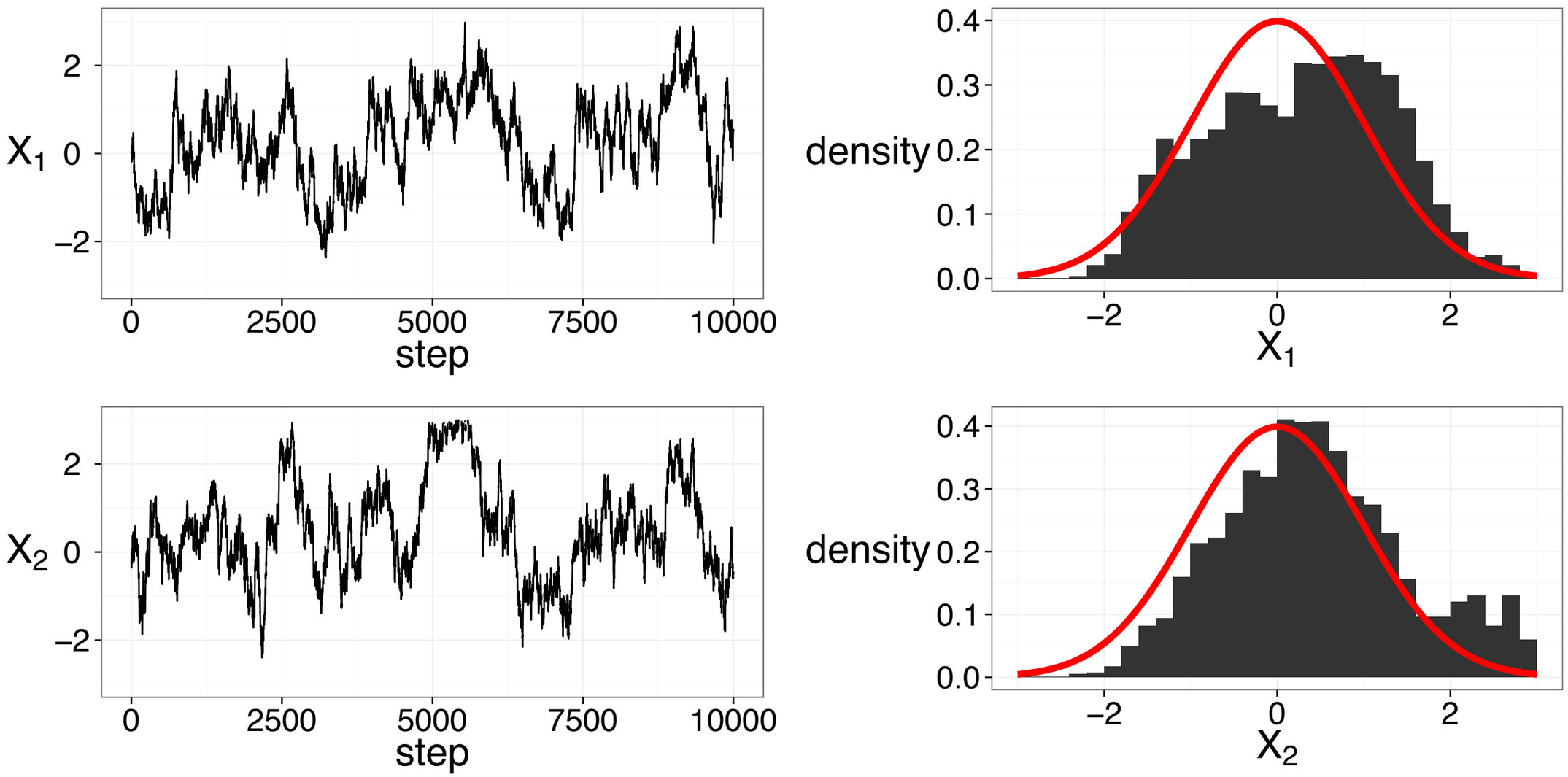
- Suppose the target distributions over $\mathbb{R}^2$

$$\pi(x) = \mathcal{N}\left(x; \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}\right)$$

- Suppose we use a RWM proposal with covariance $\sigma^2 I$

$$\eta(\varepsilon) = \mathcal{N}\left(\varepsilon; 0, \sigma^2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right).$$

- What is the optimal choice of σ?

RANDOM WALK METROPOLIS

- RMW on a bivariate Gaussian target with $\sigma = 0.1$ and acceptance rate $\alpha \approx 0.94$